Response to Reviewers

NPH-MS-2026-57029

Guard cell size and initial conductance influence stomatal closure kinetics

Muir & Lim

We thank the editor and both reviewers for their thoughtful and constructive comments. We have thoroughly responded to all reviewer comments through additional analyses that resulted in some important changes to our conclusions. The revised manuscript includes three new supplementary notes, 11 new supplementary figures, and one additional table. The reviewer comments appear below in boxes, followed by our responses in blue. To help readers easily identify the most important changes, we have included many of the new figures and substantive text changes alongside the responses.

Reviewer 1

Comment R1.1 — Realized VPD as a covariate

> It seems like the realized VPD stimulus reported varies systematically among treatments and this might needs to be treated as part of the analysis rather than as a supplementary caveat, as in Lines 238–249 are unusually candid and useful, as the authors state that scrubbing incoming water vapor "did not standardize chamber air humidity because leaf transpiration introduced water vapor," and that RH was systematically higher at high measurement light, in sun grown plants and in amphistomatous measurements because g_{sw} was greater in those conditions. Table S3 then shows that the mean VPD during measurements ranged from 2.50 kPa in sun/high/amphi curves to 2.92 kPa in shade/low/pseudohypo curves. I think this matters because the actual forcing variable in the experiment is not simply the experimental treatment label, but also is the VPD trajectory experienced by the leaf. If leaves with higher conductance both have higher f_{gmax} and also humidify the chamber more strongly, then f_{gmax} is partly entangled with the realized stimulus. I would love to see the authors include curve level VPD covariates in the model such as median VPD during the fitted interval, final VPD if available, VPD step magnitude, and perhaps the rate at which VPD changed. At minimum, should show whether the $f_{\mathrm{gmax}} \rightarrow \tau$ effect remains after accounting for actual VPD exposure, and ideally they should run a matched VPD sensitivity analysis excluding curves outside a narrow VPD range. I would be careful describing the VPD treatment as though the desiccant step created a common perturbation across treatments unless this is demonstrated. This distinction is important because VPD is

a primary driver of stomatal conductance and plant water relations, and the physiologically relevant stimulus is the leaf to air evaporative gradient not simply the inlet humidity setting.

Response:

We added Notes S1 to the Supporting Information to address the potential for variable VPD stimuli to confound our results, and updated the Materials and Methods to point to it. Note that in response to Comment R1.7, we found that g_{i} is more strongly associated with τ than f_{gmax}. Hence, we discuss whether effects g_{i} and l_{gc} on τ are robust once variation in VPD is incorporated into the analysis. We find that both effects are robust but slightly weaker once final VPD is included as a covariate in the selected model. The fixed effect of g_{i} on $\log(\tau)$ (original: 0.32 [0.23, 0.41]; VPD-adjusted: 0.27 [0.18, 0.36]) and the phylogenetic correlation between l_{gc} and τ (original: 0.59 [0.14, 0.88]; VPD-adjusted: 0.54 [-0.04, 0.88]) were both slightly weaker after adding final VPD as a covariate of τ and λ (Fig. R1; Figure S4 in the Supporting Information). One caveat is that we do not have data on the initial rate of change in VPD, which limits our ability to evaluate whether this influenced stomatal kinetic responses; we added a recommendation for future research on this point at the end of Notes S1.

We extracted the leaf-to-air VPD trajectory for every curve from the logged humidity data and summarized it with two curve-level covariates: the VPD at the end of the fitted interval ("final VPD") and the rate at which VPD approached that final value ("VPD saturation rate"). We considered adding median VPD and VPD step magnitude as well, as the reviewer suggested. However, median VPD and final VPD are highly correlated ($r = 0.93$; Fig. R2 a; Figure S1 in the Supporting Information), so we chose to use the final VPD. Final VPD contains the same information as the VPD step magnitude since all leaves were started at the same RH and Tleaf, hence starting VPD was approximately constant.

We attempted to incorporate the rate of VPD change as a covariate, but determined this was not useful. Unfortunately, we did not log data during the WWR when the initial step change in VPD occurred. We instead fit a saturating exponential, $\mathrm{VPD}(t) = B + (R_0 - B)\exp(-kt)$, to each curve's VPD trajectory (median $R^2 = 0.99$) and use the rate constant k ("VPD saturation rate"). This rate constant is only weakly correlated with final VPD ($r = 0.31$; Fig. R2 c). However, k is strongly correlated with τ ($r = 0.77$ on a log-log scale; Fig. R3) because it is mostly caused by decreasing g_{sw}. We can see this clearly by comparing the rate of change in the water vapor concentration of the reference airstream, upstream of the leaf (W_{r}), versus that of the sample airstream, which reflects humidity inside the chamber (W_{s}). If the change in VPD is caused by g_{sw}, we expect a positive correlation between the rate of VPD change and the rate of W_{s} change, which is exactly what we see (Fig. R4 a; Figure S3 in the Supporting Information). In contrast, the rate of W_{r} change is uncorrelated with the rate of VPD change (Fig. R4 b; Figure S3 in the Supporting Information). Hence, it would not be statistically valid to include the rate of VPD change as a covariate in predicting τ since both are caused by the same underlying variable, the rate of change in g_{sw}.

To directly test whether the realized VPD stimulus confounds the g_{i} and l_{gc} effects on τ, we refit our selected model with final VPD added as a covariate of τ and λ. As reported above, the g_{i} and l_{gc} effects were slightly weaker when final VPD was included as a covariate (Fig. R1). Final VPD itself had a negative fixed effect on $\log \tau$ (-0.41 [-0.53, -0.27]), consistent with higher realized VPD accelerating stomatal closure.

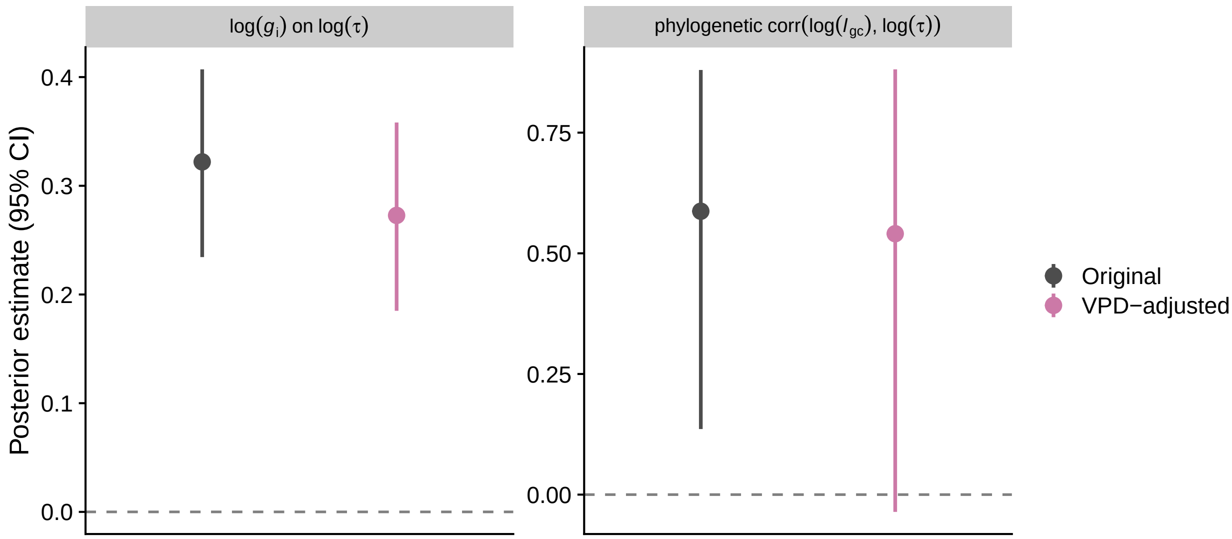


Fig. R1: The g_{i} effect on τ and the phylogenetic correlation between l_{gc} and τ are robust to accounting for final VPD. Posterior means and 95% credible intervals for the fixed effect of $\log(g_{\mathrm{i}})$ on $\log(\tau)$ (left) and the phylogenetic correlation between $\log(l_{\mathrm{gc}})$ and $\log(\tau)$ (right), comparing the original model (grey) to a model refit with final VPD added as a covariate (pink).

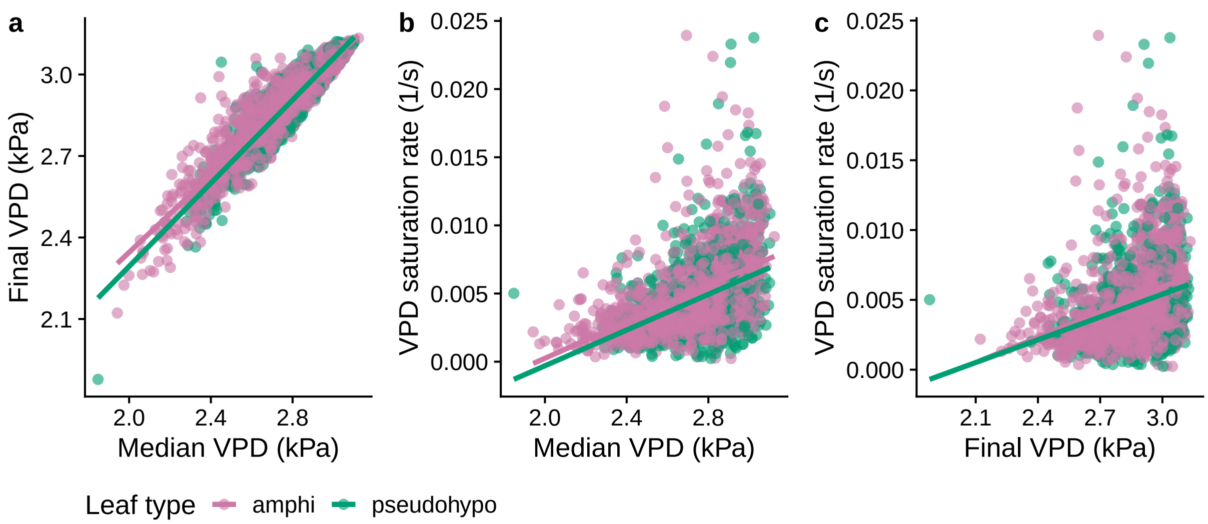


Fig. R2: Pairwise relationships among curve-level VPD covariates: (a) median vs. final VPD, (b) median VPD vs. VPD saturation rate, (c) final VPD vs. VPD saturation rate. Each point is one humidity response curve, colored by leaf type; lines are ordinary least-squares fits within leaf type. VPD saturation rate is the rate constant of a saturating exponential fit to the VPD trajectory during the fitted interval.

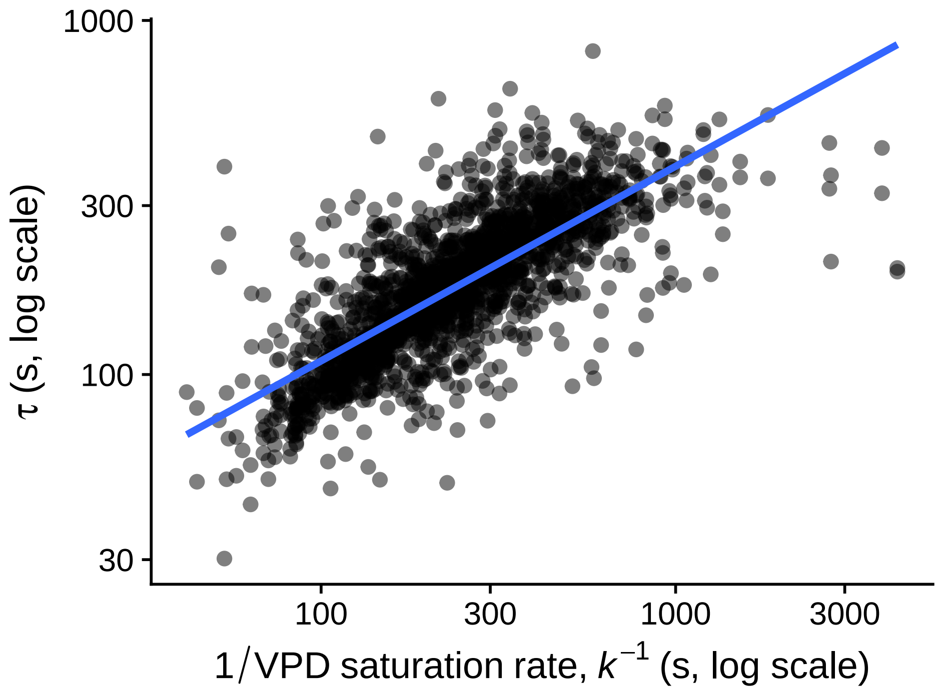


Fig. R3: The stomatal closure time constant τ is positively correlated with the inverse of the VPD saturation rate, k^{-1} (both log scale; $r = 0.77$), consistent with k being largely driven by the same g_{sw}-decline signal that determines τ, rather than an independent property of the VPD stimulus. Each point is one humidity response curve.

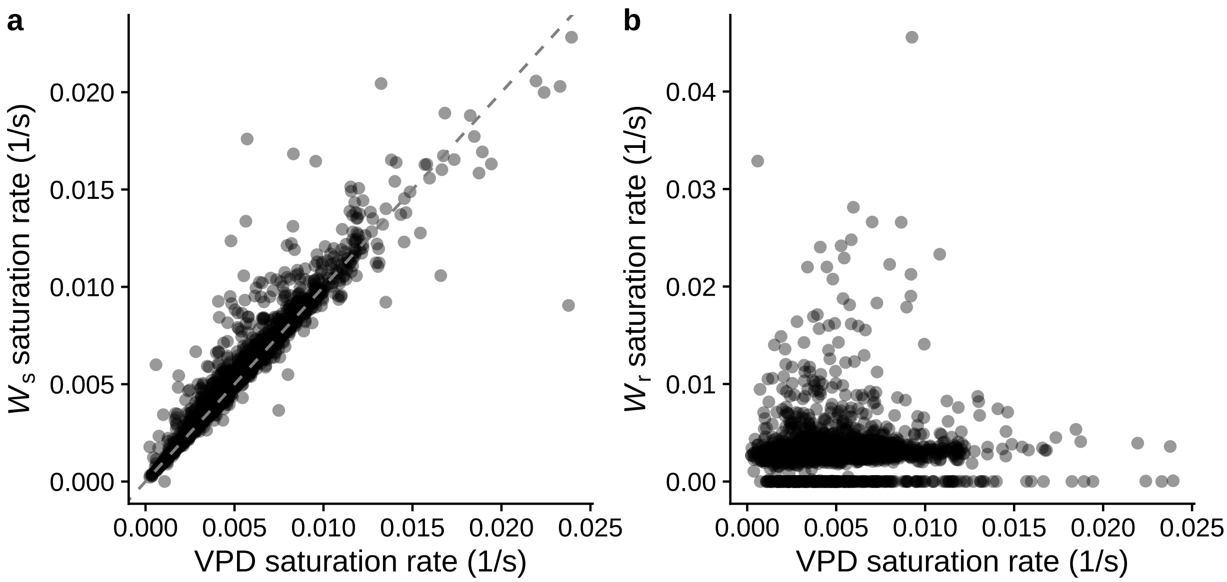


Fig. R4: The VPD saturation rate is driven by the sample airstream, not the reference airstream. (a) VPD saturation rate vs. the saturation rate of W_s (sample cell water vapor concentration, which has equilibrated with the leaf); the dashed line is the 1:1 line. (b) VPD saturation rate vs. the saturation rate of W_r (reference cell water vapor concentration, upstream of the leaf). Each point is one humidity response curve; rate constants come from the same saturating-exponential model used for VPD.

Comment R1.2 — Mislabeled VPD summary statistic

Related to the previous point, the manuscript appears to call the Table S3 values "final RH" and "final VPD," but I found the code and caption indicate something different. At lines 245 to 247, the text says that "the final RH" and "VPD" values in Table S3 elicited rapid stomatal response. However, the Table S3 caption says the values are calculated as the median RH and VPD across time points within each curve and then averaged among curves. The code confirms this: `r/09_make-tbl-vpd.R`, lines 7–14, calculates median RH and VPD per curve, and lines 32–40 average those medians across treatment combinations. That is not the same as final RH/VPD nor necessarily the VPD step magnitude. It would be great if author can correct the wording and, more importantly, to report the full VPD trajectory more transparently, with pre-step VPD, VPD at the start of logging, median VPD during the fitted RWR, final VPD, and ideally the change in VPD over time, because the paper interprets variation in τ as intrinsic closure kinetics, while the stimulus itself may vary across the same treatments used to explain τ.

Response:

We thank the reviewer for catching the discrepancy between what we reported in Table S3 versus the actual content. As noted in the response to the previous comment, the median and final RH and VPD are highly correlated, so this mistake did not affect the interpretation. We revised Table S3 (now Table S2) to report the average RH and VPD at three time points: when logging started, when logging finished, and the median during the logging interval. We also added the curve-level values for these variables to Table S1. We did not record the pre-step VPD but it was nearly identical for every leaf because we set leaf temperature to 25 °C and RH to 70% to standardize starting conditions. This is equivalent to VPD ≈ 0.82 kPa, as mentioned in the Materials and Methods. As mentioned in the previous response, we also do not have data on the initial rate of VPD increase.

Comment R1.3 — f_{gmax} is not an independent aperture measurement

I like f_{gmax} as a quite intriguing predictor but it is not an independent direct measure of stomatal aperture. The manuscript motivates f_{gmax} as proportional to aperture relative to maximum aperture at lines 115–133, defines it as g_i/g_{max} at lines 273–279 and then interprets higher f_{gmax} as stomata being more open at lines 397–404 and as a mechanistic driver of slower closure at lines 506–513. I think the hypothesis is interesting, but the current evidence is not enough to treat f_{gmax} as a direct aperture measurement. The numerator, g_{i}, is estimated from the same nonlinear curve that also gives τ and λ, so parameter covariance in the Weibull fit could generate an association between g_{i} and τ even if the underlying biological relationship is weaker. The denominator g_{max} is also model-derived from stomatal density, guard cell length, assumed pore geometry, and conductance calculations. I think one paper by the Sack and Buckley group explicitly describes anatomical g_{max} as a theoretical estimate of maximum stomatal diffusion capacity that cannot usually be reached in practice, which is useful but not identical to observed aperture. I would ask the authors to validate or reframe the variable. Specifically, they should report posterior correlations between fitted g_i and τ within individual curves, run a null simulation showing whether the curve fitting procedure alone can induce a g_i–τ association, and repeat the main analysis using alternative predictors such as absolute g_i, observed maximum g_sw, g_max, and g_i and g_max as separate predictors. If no direct pore aperture measurements exist, I would also suggest using more cautious language such as "conductance fraction of anatomical maximum" rather than "aperture."

Response:

We thank the reviewer for raising this point because it revealed a mistake in our explanation of the methods that is also relevant to Comment R1.4 below. Figure 1a and parts of the written description incorrectly stated that the fitted interval began as soon as the RWR began. To clarify, we deliberately began logging each curve once g_sw had declined back down to its pre-step value. We did this by recording the pre-step g_sw and starting the log protocol when g_sw returned to value after the WWR. The reason for this protocol was that preliminary experiments showed the relationship between g_sw and A analyzed in Muir *et al.* (2025) was "off" over the interval of g_sw between its maximum and when it returned to the pre-step value (the interval between the start of the RWR and when we started logging as depicted in Figure 1a). Our interpretation is that during the WWR g_sw increases faster than photochemistry can be upregulated, such that A doesn't increase as much as you would expect for that amount of change in g_sw. This point is tangential to our analysis of stomatal kinetics, but we hope it clarifies the rationale. Hence, we only have a single estimate of f_gmax where g_i is the pre-step g_sw, as depicted in the revised version of Figure 1a. We do not have data on maximum g_sw at the end of the WWR. g_i is therefore also the maximum observed g_sw because we allowed leaves to acclimate under conditions (high light, low VPD) conducive to high g_sw. In principle, if the relationship between guard cell turgor and g_sw (Figure 1d) within a leaf is fixed then we should be able to estimate the effect of f_gmax on τ regardless of where along the response curve we start, but we cannot evaluate this in more detail with the data we have.

We revised Figure 1a and the Materials and Methods (lines 255-256) to describe this precisely, and moved the "Log data" marker in the figure to the point where g_sw returns to g_i, distinct from the start of the RWR itself, which begins where g_sw first starts to decline from its WWR peak. We also added a caveat to the Notes S1 on lines 930-936 noting this limitation.

We also added simulations demonstrating that the association between g_i, f_gmax, and τ is unlikely to be an artifact of simultaneously estimating g_i and τ (see Notes S2). For every real curve, we simulated a synthetic g_sw trajectory using that curve's own measurement times, residual variance, and fitted τ and λ values, but with fitted values of g_i and g_f randomly swapped with another curve. Hence, synthetic curves retain the observed associations between τ and λ from curve i and the observed associations between g_i and g_f from curve j. Since i and j are chosen at random, there is no actual association between g_i and kinetic parameters among the synthetic curves. Since the denominator in f_gmax is estimated independently from stomatal anatomy, it is sufficient to evaluate a spurious association between g_i and τ. For each synthetic curve, we re-fit the same Weibull functional form to each simulated curve (via `nls()`, validated against the full Bayesian pipeline used for the real data: $\mathrm{cor}(\tau) = 0.99$, $\mathrm{cor}(\lambda) = 1.00$) and computed the correlation between the *estimated* g_i and $\log(\tau)$. We repeated this 1,000 times to build a null distribution of this correlation.

The null-simulation correlations ranged from -0.06 to 0.09 (median -3.1×10^{-3}) across the 1,000 synthetic curves – indistinguishable from zero – whereas the real, observed correlation between $\log(g_\mathrm{i})$ and $\log(\tau)$ is 0.39 [0.35, 0.42] (Fig. R5; Figure S13 in the Supporting Information), far outside the null range (empirical $P < 0.001$). This indicates that the curve-fitting procedure alone cannot explain the observed g_i-τ association, and that it more plausibly reflects a real relationship in the data.

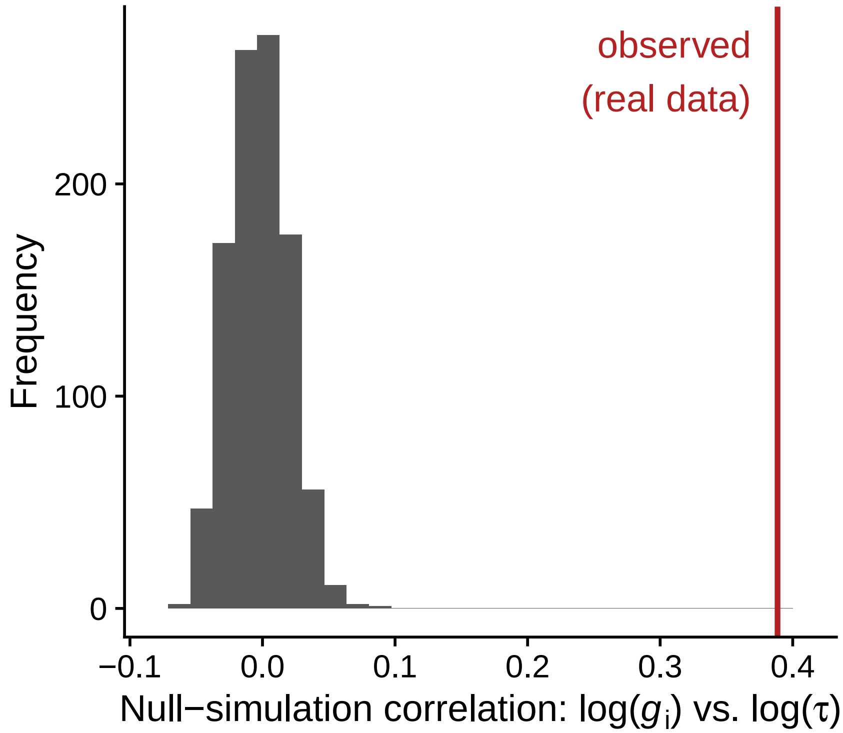


Fig. R5: Null distribution of the correlation between estimated g_i and estimated τ across 1,000 simulated datasets in which the true g_i and g_f are fixed at their real median values for every curve (so g_i has no true relationship with τ by construction), compared to the observed correlation in the real data (red line).

While g_{max} is theoretical to some extent, it does seem to be a good description of maximum g_{sw} in experimentally controlled *Arabidopsis* plants (Dow *et al.*, 2014). We added this in the Materials and Methods on lines 293-295.

To address the last part of the reviewer's comment, we substantially revised the model set we considered and this led to significant changes in our conclusions. The reviewer is correct that we do not have direct estimates of stomatal aperture and we revised the manuscript to reflect that our inferences are indirect. We also tested for independent effects of g_i and g_{max} that are described in response to Comment R1.7 below. The new approach revealed that g_i is more closely associated with kinetic parameters than g_{max}. Therefore, both greater density and/or aperture may slow the rate of closure. We have revised the Results and Discussion in several places to reflect this change in our interpretation. The main change in our interpretation is stated in this paragraph (lines 564-577):

> "The association between g_i and τ could be explained by the hypothesis that guard cell wall elasticity declines at greater turgor pressure (Franks *et al.*, 2012), like a rubber band that is stretched under tension. However, if this were the case, we would also expect a negative association between g_{max} and τ because leaves with greater g_{max} could have the same g_i with a smaller stomatal aperture. However, this effect was weakly

supported (Figures S10 and S12). The selected model did not include an individual-level effect of g_{max} on kinetic parameters. Although the second-best model included g_{max} and the effect was in the predicted direction, the effect was not significantly different than 0 (Figure S7). The weak effect of g_{max} compared to g_{i} casts doubt on whether a nonlinear relationship between turgor pressure and pore aperture is the best explanation. An alternative, albeit not mutually exclusive explanation, is a nonlinear relationship between pore aperture and g_{sw} that weakens the association between these variables at high g_{sw}. Airspace resistance within the substomatal cavity and stomatal interference are plausible mechanisms that would have this effect (Ting & Loomis, 1965; Kaiser, 2009; Lehmann & Or, 2015), although it is not clear that they are sufficient to produce the observed relationship between g_{i} and τ in wild tomatoes."

Comment R1.4 — Timing of g_{i} is ambiguous

> I also found the timing of g_{i} is ambiguous because the authors start logging after the wrong way response. Figure 1A and lines 41–45 correctly distinguish the WWR and RWR, but lines 248–249 state that data logging began after the transient WWR. Then lines 273–278 say that fitted g_{i} is a good proxy for conductance "at the time of the step change in humidity." These statements do not seem consistent. If logging begins after the WWR, then g_{i} is the initial conductance of the analyzed RWR segment and therefore not necessarily conductance at the humidity step. Figure 1A actually illustrates this distinction nicely with the VPD step occurring before the orange "log data" interval. It would be great if author define g_{i} throughout as the fitted initial conductance at the start of the analyzed RWR, unless pre-step conductance was recorded and used. If pre-step steady state g_{sw} exists, the authors could also consider computing a pre-step f_{gmax} and test whether it predicts τ, as that would be much more convincing mechanistically than using a parameter estimated from the RWR itself.

Response:

Please see response to Comment R1.3 above.

Comment R1.5 — High vs. low light confounded with measurement order

> Another confusing point is that the current high versus low measurement light comparison is confounded with measurement order and prior stress history. Lines 224–226 say that high light curves were always measured first, and lines 250–251 say the leaf was then acclimated to low light and high RH before inducing the low light closure curve. This means the largest treatment effect in the paper is also a fixed sequence effect with high light always first, and low light always following a prior high light, low humidity closure event on the same leaf. That matters because high measurement light strongly increases both f_{gmax} and τ and the mediation analysis then interprets part of the light effect as operating through f_{gmax}. I do not see how the current design separates measurement light from order, recovery state, prior VPD exposure, elapsed time, or

time of day. It might be better to call this a "measurement light/order condition" unless they can show that sequence effects are negligible.

Response:

We chose this design because the primary goals of the study were to compare amphi and pseudohypo curves on the same leaf within the same treatment to estimate "amphistomy advantage" (Muir *et al.*, 2025). Preliminary studies indicated the order did not affect amphistomy advantage but that it was much faster to go from the high light to low light compared to the reverse. Therefore, we opted to increase our throughput and sample size at the potential cost of some confounding between measurement order and treatment effect. Unfortunately, we did not observe whether the measurement order affected stomatal kinetics. Hence, we cannot say for sure whether order, recovery state, or prior VPD exposure confounds measurement light intensity. We controlled for time of day by running curves throughout the day and randomizing the order among accessions and growth light treatments. However, we think the bias induced by measurement order is likely small compared to the very large effect of measurement light intensity on g_i (+77% for shade plants; +139% for sun plants):

Table 1: Average g_i in different treatment combinations across the amphi leaf type curves.

Growth light	Measurement: low	Measurement: high
shade	0.209	0.370
sun	0.265	0.634

We have already mentioned that this analysis is opportunistic and therefore some of the experimental design choices are sub-optimal for the purpose of studying stomatal kinetics. We addressed the reviewer's concerns about measurement order more specifically by:

1. Mentioning the potential for measurement order to confound effects of measurement light treatments (Materials and Methods lines 225-228).

   > "As a consequence, measurement light intensity is confounded with measurement order in our design: low light-intensity curves were always measured after high light-intensity curves on the same leaf, following a low-humidity closure event, so we cannot fully separate an effect of light intensity per se from order, recovery state, or prior VPD exposure (see Discussion)."

2. Advising caution in interpreting effects of measurement light treatment and mediation analysis (Discussion lines 592-599).

   > "The comparison between measurement light intensity treatments should be interpreted cautiously, however, because it was confounded with measurement order: high light curves always preceded low light curves on the same leaf, following a prior low-humidity closure event (see Materials and Methods). We cannot rule out that some of the apparent effect of measurement light intensity on g_i and τ, and the corresponding indirect path through g_i in our mediation analysis (Figure 7), reflects order, incomplete recovery, or prior VPD exposure rather than measurement light intensity itself. This concern applies specifically to the measurement light comparison and not to growth light intensity or leaf type, which were randomized or alternated, respectively."

3. Adding that the average time between high and low measurement light intensity curves was 48.9 minutes (line 258).

   "...on average 48.9 minutes after the start of the high-light curve."

Comment R1.6 — "Causal mediation" language too strong

> Lines 339–347 describe "causal mediation analysis," and lines 445–455 report mediated effects and Figure 7 presents path diagrams implying treatment effects flowing through f_{gmax} to τ. I understand why the authors did this but the causal assumptions are too strong for the current design. f_{gmax} is not randomized, it is estimated from the same curve as τ, it is affected by actual VPD trajectory and measurement order and it is likely related to unmeasured hydraulic state. In the standard mediation literature, causal mediation requires strong assumptions about mediator–outcome confounding usually framed as sequential ignorability and sensitivity analysis is a central part of the approach. It could be great if authors replace "causal mediation analysis" with "statistical decomposition" or "path analysis" unless they add a DAG and sensitivity analysis. The text could still say that treatment-associated changes in τ are statistically consistent with an indirect path through f_{gmax}, but I would avoid saying that f_{gmax} "mediates plasticity" as a causal conclusion.

Response:

We agree and have replaced "causal mediation analysis" with "path analysis" throughout the manuscript. Note that in response to Comment R1.3 and R1.7, we found stronger evidence for g_{i} rather than f_{gmax} affecting τ. Therefore, we reran path analysis using g_{i} in place of f_{gmax} as described in the revised "Path analysis to test for plastic effects" section. We no longer say that g_{i} "mediates plasticity" as a causal conclusion. We now state only that treatment-associated changes in τ are statistically consistent with an indirect path through g_{i}, alongside a direct path, exactly as the reviewer suggested.

We also drew the DAG the reviewer requested (Fig. R6; Figure S14 in the manuscript), showing the assumed causal structure (treatment $\to g_{\text{i}} \to \tau$ and treatment $\to \tau$ directly) alongside an unmeasured confounder U representing exactly the kind of variable the reviewer raised (hydraulic status, measurement order, realized VPD trajectory) that would violate sequential ignorability if it affected both g_{i} and τ.

To directly address sequential ignorability, we ran a sensitivity analysis in Notes S3 following the general framework of Imai *et al.* (2010), parameterized by $\rho = \text{Cor}(e_{g_{\text{i}}}, e_\tau)$, the residual correlation between the g_{i} and τ equations. Under a violation of sequential ignorability with residual correlation ρ, the bias-adjusted mediator-on-outcome coefficient is $\beta_2(\rho) = \hat{\beta}_2 - \rho \cdot \sigma_\tau / \sigma_{g_{\text{i}}}$, so the average causal mediation effect (ACME) is $\text{ACME}(\rho) = \gamma_1 \cdot \beta_2(\rho)$, and the "breakdown point" ρ^* at which the ACME is exactly zero is $\rho^* = \hat{\beta}_2 \cdot \sigma_{g_{\text{i}}} / \sigma_\tau$. In these equations, β_2 is the effect of g_{i} on τ and γ_1 is the effect of treatments on g_{i}.

Our selected model already estimates the residual correlation between the g_{i} and τ equations ($\rho_{\text{estimated}}$, from `set_rescor(TRUE)`), because we fit all five responses (l_{gc}, g_{i}, g_{max}, τ, λ) jointly. We treat $\hat{\beta}_2$ as the "naive" coefficient in the sensitivity framework above, and the model's own estimated residual correlation as an estimate of ρ.

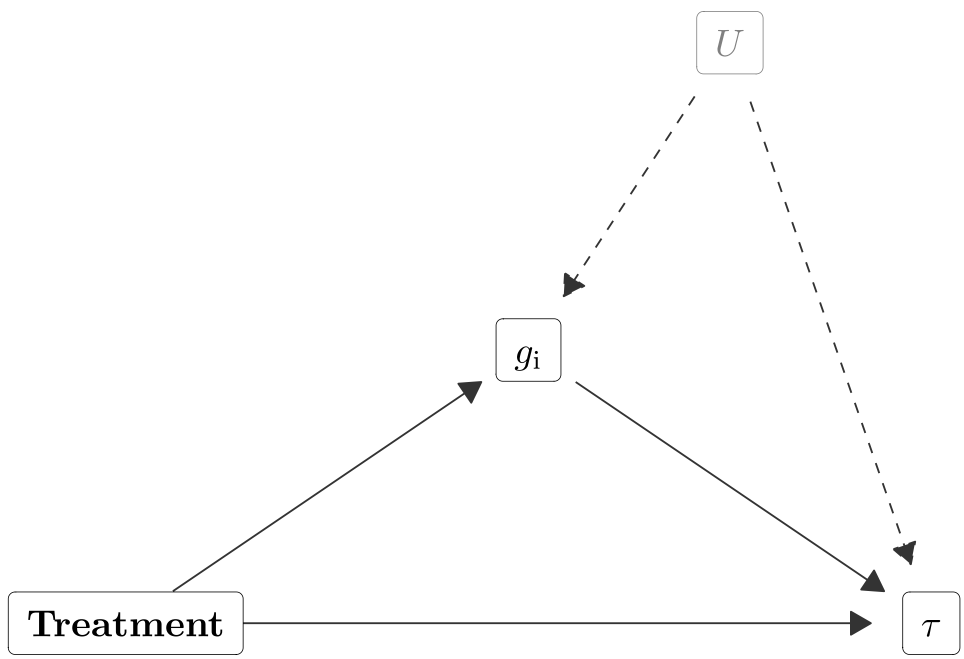


Fig. R6: Assumed causal structure of the g_{i}-τ path analysis. Treatment has a direct effect on τ (bottom arrow) and an indirect effect acting through g_{i} (upper path). An unmeasured confounder U (dashed arrows) affecting both g_{i} and τ would violate the sequential ignorability assumption required to interpret the indirect path causally.

We found $\rho_{\text{estimated}} = 0.06$ [-0.07, 0.18] and $\rho^* = 0.40$ [0.29, 0.51] (Fig. R7, Figure S15 in the revised manuscript). These posteriors do not substantially overlap and the posterior probability that the estimated residual correlation is smaller in magnitude than the breakdown point is nearly 100%. In other words, the plastic mediation effects of g_i on τ are probably not explained by unmeasured confounders. Nevertheless, we took the reviewer's suggestion to avoid strong causal claims because the mediator variable g_i was not experimentally controlled and could plausibly be related to τ through confounding variables, though our analysis does not support this conclusion.

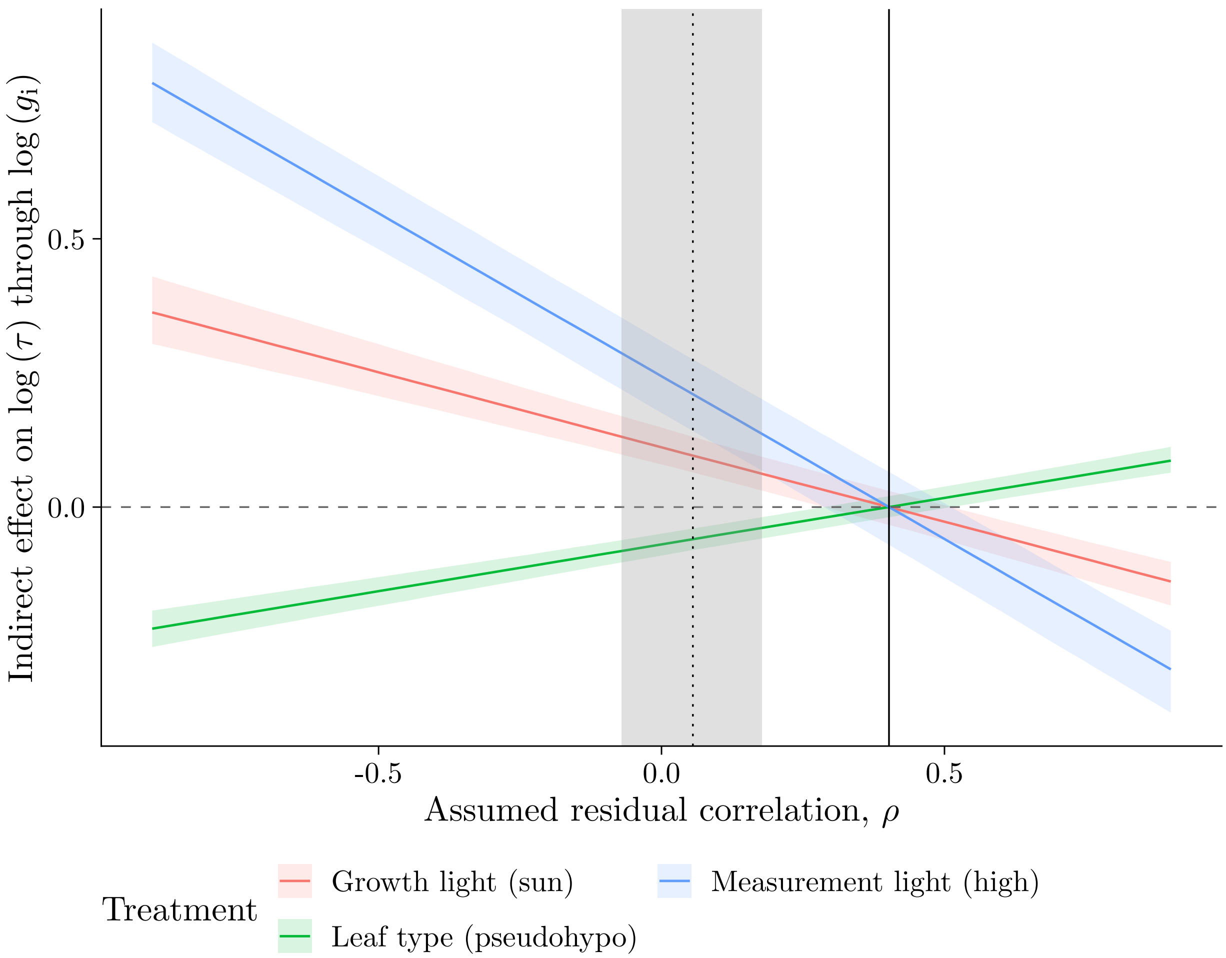


Fig. R7: **Sensitivity of the indirect effect of each treatment on $\log \tau$, acting through g_i, to an assumed residual correlation ρ between the g_i and τ equations.** The vertical dotted line is the model's own estimated residual correlation ($\rho_{\text{estimated}}$, shaded band = 95% credible interval); the vertical solid line is the breakdown point ρ^* at which the indirect effect is zero.

Comment R1.7 — Model selection is more subjective than admitted

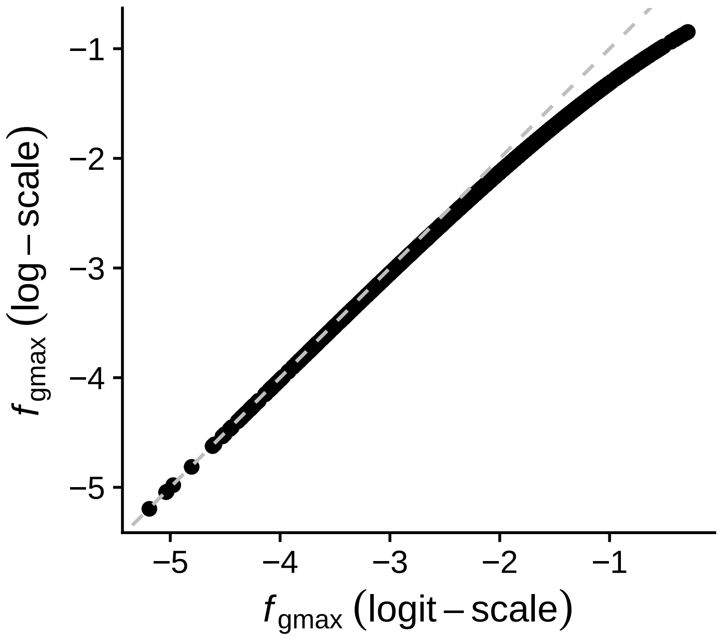
I also found the model selection is more subjective than the present manuscript currently admits. Lines 332–337 say that models were compared using LOOIC and models within two ΔLOOIC standard errors were considered plausible. Table 2 shows six plausible models and the selected model is not the minimum LOOIC model. Lines 430–435 then acknowledge that judgment calls were used, which I appreciate. However, the code is more explicit than the manuscript: `r/11_compare-models.R`, lines 74–84, says the top four model order changed across reruns and that model 6 was chosen because it was "clearest to interpret." That may be defensible but it should be stated plainly in the manuscript. It would be great if authors can present focal estimates across all plausible models not only the selected model, especially for $l_{\mathrm{gc}} \to \tau$, $f_{\mathrm{gmax}} \to \tau$, $l_{\mathrm{gc}} \to \lambda$, and $f_{\mathrm{gmax}} \to \lambda$. They could also consider reporting Pareto $\hat{k}$ diagnostics and Monte Carlo uncertainty for PSIS-LOO, because PSIS-LOO model comparison relies on those diagnostics. If you read the `loo` documentation where it emphasizes Pareto-k, PSIS effective sample size and MCSE as reliability checks.

Response:

The phrase "clearest to interpret" was informal shorthand (CDM) for the fuller explanation in the text, so there is no discrepancy to correct. To make model selection less subjective, we now describe explicit criteria in the revised "Model selection and parameter estimation of individual-level effects" section. We first narrowed the candidate set using LOOIC, then evaluated the remaining models on the focal parameters addressing our hypotheses — individual-level and phylogenetic effects of anatomical traits on kinetic traits. Among these, we selected the model with the highest proportion of significant focal parameters (95% CIs excluding 0), regardless of whether the direction matched our hypotheses. This criterion favors more parsimonious models, particularly ones without additional non-significant terms, and makes the selected model easier to interpret. This criterion is somewhat arbitrary and not based on formal statistical inference. We therefore also discuss alternative interpretations supported by other plausible models. In the revised model comparison, this approach happened to select the model with the lowest LOOIC (see Table S4), but that would not always be the case.

To address the second half of the comment, we changed the candidate set of models as alluded to in response to Comment R1.3. In the original manuscript, we tested individual-level and phylogenetic effects of the composite variable f_{gmax} on a logit-scale. We realized we could achieve the same goals by modeling the effects of component traits (g_{i} and g_{max}) on kinetic traits directly. We reasoned that if f_{gmax} *per se* affects kinetics, then both g_{i} and g_{max} should be retained in candidate models and have significant effects on kinetic traits. Conversely, if only one of the component traits affects kinetics, then it should be retained in the selected model, while the other is dropped. Our new approach is described in the "Model selection and parameter estimation of individual-level effects" section.

One minor discrepancy between the former and current approaches is that the current approach models the f_{gmax} on a log rather than logit-scale, since $\log\left(f_{\mathrm{gmax}}\right) = \log\left(g_{\mathrm{i}}\right) - \log\left(g_{\mathrm{max}}\right)$. However, the two transformations of f_{gmax} are highly correlated in our data:



The new modeling approach changed our interpretation somewhat and these changes are reflected in the revised Results and Discussion (see example above in response to Comment R1.3). Overall, individual-level variation in g_{i} explained much more of the variation in kinetics than g_{max}. We therefore revised the figures and mediation analyses to emphasize the role of g_{i} rather than f_{gmax}. We also note that the second-best model retained g_{max}; the estimated coefficient was in the predicted direction, but it was not significant.

In response to the request to present focal estimates across all plausible models, Fig. R8 and Fig. R9 show individual-level fixed effects and phylogenetic correlations, respectively, of l_{gc}, g_{i}, and g_{max} on both τ and λ across all six plausible models identified by LOOIC (Figures S7-S8 in the Supporting Information). Estimates are qualitatively consistent across the plausible model set, supporting our decision to base inference on the selected model while also discussing alternative interpretations from the other plausible models where relevant.

Finally, we report Pareto $\hat{k}$, PSIS effective sample size (ESS), and Monte Carlo standard error (MCSE) diagnostics for the LOOIC comparison, as the reviewer requested (`r/38_get-loo-diagnostics.R`; Table S5 in the Supporting Information). Across the six plausible models, the vast majority of the 2125 curves had Pareto $\hat{k} \leq 0.7$ rather than "very bad" range; no model had any curve with $\hat{k} \geq 1$. The minimum PSIS ESS across curves was as low as 94 for some models, which is well below the conventional rule-of-thumb of a few hundred effective draws, so the corresponding PSIS-LOO estimates for those individual curves (and the resulting MCSE for the model as a whole) should be treated with some caution rather than taken at face value. We have added this caveat to the Materials and Methods (lines 364-368).

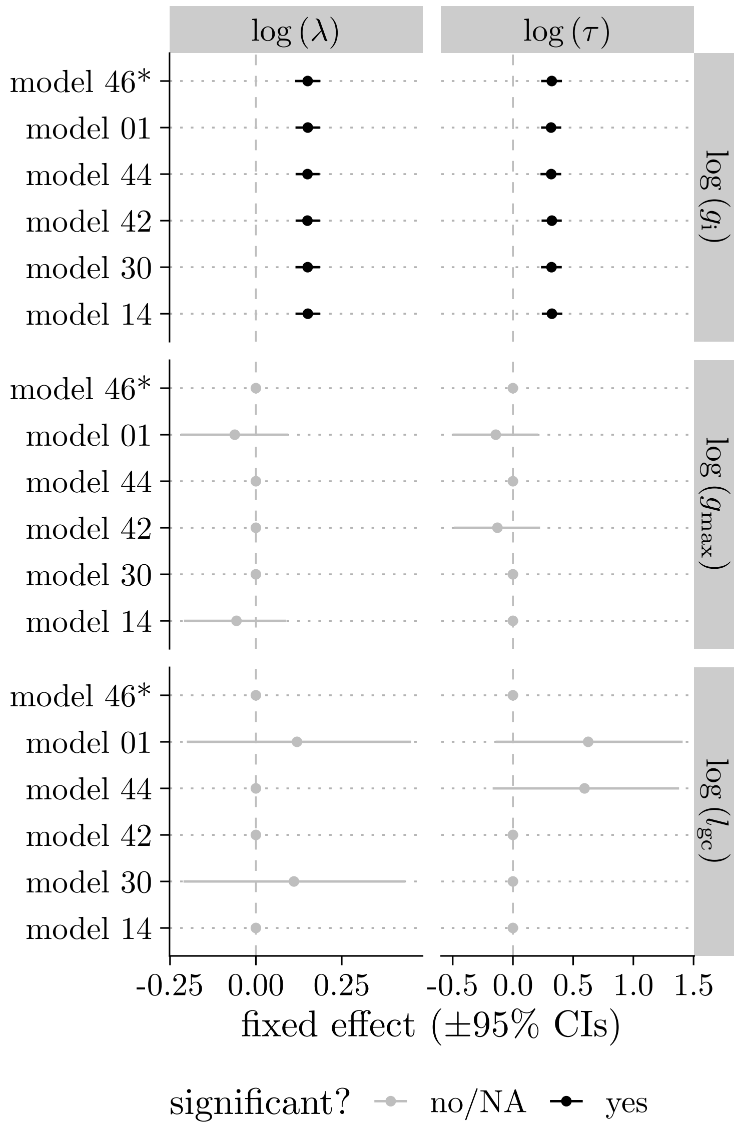


Fig. R8: Individual-level (fixed-effect) estimates of guard cell length (l_{gc}), initial stomatal conductance (g_i), and anatomical maximum stomatal conductance (g_{max}) on the stomatal closure time constant (τ) and lag time (λ), across all six plausible models identified by LOOIC (Table S4 in the manuscript). Points and horizontal bars are posterior medians and 95% credible intervals; the selected model is marked with an asterisk (*). Black indicates estimates whose 95% credible interval excludes zero; grey indicates intervals that overlap zero.

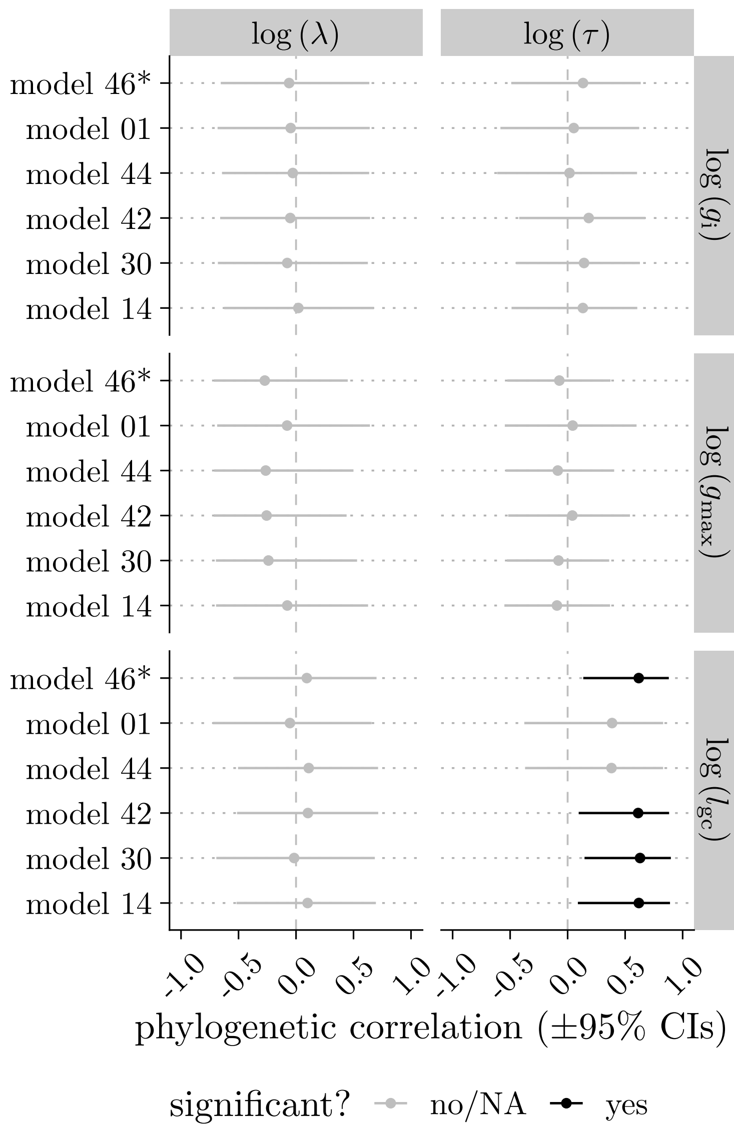


Fig. R9: Phylogenetic correlations between guard cell length (l_{gc}), initial stomatal conductance (g_i), and anatomical maximum stomatal conductance (g_{max}) and the stomatal closure time constant (τ) and lag time (λ), across all six plausible models identified by LOOIC (Table S4 in the manuscript). Points and horizontal bars are posterior medians and 95% credible intervals of the phylogenetic correlation; the selected model is marked with an asterisk (*). Black indicates estimates whose 95% credible interval excludes zero; grey indicates intervals that overlap zero.

Table 2: Pareto $\hat{k}$, PSIS effective sample size (ESS), and Monte Carlo standard error (MCSE) diagnostics for the six plausible models identified by LOOIC (Table S4 in the manuscript). The selected model is in bold. Good/bad/very bad refer to the conventional Pareto $\hat{k}$ thresholds of 0.7 and 1. MCSE($\mathrm{elpd}_{\mathrm{loo}}$) is left blank (‘–’) for models with at least one curve with $\hat{k} \geq 1$, for which the MCSE estimate is undefined.

Model	ΔLOOIC	SE	n curves	Good $\hat{k}$ (%)	Bad $\hat{k}$ (%)	Very bad $\hat{k}$ (%)	Min. PSIS ESS	MCSE($\mathrm{elpd}_{\mathrm{loo}}$)
fit_46	0.00	0.00	2125	99.95	0.05	0.00	155	0.99
fit_01	3.58	2.95	2125	99.95	0.05	0.00	149	0.99
fit_14	5.20	2.72	2125	100.00	0.00	0.00	137	1.00
fit_30	4.92	2.90	2125	99.95	0.05	0.00	158	1.00
fit_42	4.91	2.92	2125	100.00	0.00	0.00	94	1.00
fit_44	4.80	2.96	2125	100.00	0.00	0.00	172	0.99

Comment R1.8 — Single-chain Bayesian diagnostics for curve-level fits

The curve-level Bayesian diagnostics are not yet convincing enough for a dataset built from 2,124 fitted nonlinear curves. Lines 285–289 state that each curve was fit using 1,000 post-warmup draws from a single chain, with Rhat < 1.05, bulk ESS > 400 and fewer than 10 divergent transitions. The code confirms the single-chain curve fitting in `r/functions.R`, lines 8–18. I am not fully convinced with this as the only diagnostic for thousands of nonlinear curve fits especially because g_i, g_f, τ, and λ are likely correlated. Stan's diagnostics treat divergent transitions as a warning that HMC may not be fully exploring the posterior and the usual workflow relies on multiple chains to assess convergence. Perhaps no need for the authors to refit all 2,124 curves with four chains if that is computationally prohibitive, but it would be great for a stratified four-chain validation subset including difficult curves, high-τ curves, high-f_gmax curves, and low-R^2 curves. To show or supplement that the single-chain pipeline gives essentially the same τ, λ, and $g_\mathrm{i}$estimates as a more standard multi-chain fit.

Response:

We refit all curves with 4 chains instead of 1 and updated the Materials and Methods accordingly. As far as we can tell, this did not meaningfully impact any downstream analyses.

Comment R1.9 — Undocumented guard cell length outlier exclusions

I found in `r/01_join-data.R`, three guard cell length outlier individuals are removed with a comment referring to `r/scratch - gcl outliers.R`. I do not see this exclusion described in the manuscript and the referenced scratch file is not included in the package. Because guard cell length is one of the two central anatomical predictors, it would be great to see this reported. The same script also uses a first-available fallback for stomatal anatomical data when the LICOR and stomatal leaflet do not uniquely match (`r/01_join-data.R`). That may be reasonable but it could be better to document because it might add measurement error to g_max, guard cell length and therefore f_gmax.

Response:

We unintentionally neglected to explain why these three individuals were considered outliers. In the Materials and Methods (lines 271-274) we clarified that:

> "We excluded three individuals with extreme outlying guard cell length values. Preliminary analysis of variation in guard cell length among growth treatments, leaf types, and accessions identified these outliers based on a residual value greater than 0.45, which visually appeared as a clear break in the residual distribution."

Regarding the leaflet-matching fallback: when a leaflet was measured for stomatal anatomy but did not match the leaflet used for the LI-6800 humidity-response curve, we used the first available anatomical measurement from a different leaflet on the same leaf as a proxy, rather than discarding the individual. This fallback was triggered for 9 of 561 individual-by-leaf-type combinations (1.60%). We agree this could add measurement error to g_max, guard cell length, and f_gmax for those individuals, and we have added this frequency and a caveat to the Materials and Methods (line 275-278).

Comment R1.10 — Figure 1D color labeling appears reversed

In Figure 1D, the orange point is the low f_{gmax} point on the steep part of the curve and the blue point is the high f_{gmax} point on the shallow part of the curve? The figure labels themselves appear correct. However, the Figure 1 caption says the larger change occurs at low f_{gmax} "blue; steep tangent slope" and high f_{gmax} is "orange; shallow tangent slope." I found that is reversed relative to the plotted colors?

Response:

The reviewer is correct. The in-figure labels and plotted colors in Figure 1d were always consistent (low f_{gmax} = orange/steep slope, high f_{gmax} = blue/shallow slope), but we had the colors swapped in the Figure 1 caption text. We corrected the caption to match the figure.

Reviewer 2

Comment R2.1 — Abstract sentence unclear

I am not sure this sentence reads clearly: "Leaves with small guard cells and a lower g_{op} to g_{max} ratio (f_{gmax}) closed faster, but explained variation in kinetic parameters at different levels of biological organization." Perhaps it is missing 'both'? "Leaves with small guard cells and a lower g_{op} to g_{max} ratio (f_{gmax}) closed faster, but both explained variation in kinetic parameters at different levels of biological organization."

Response:

We revised this sentence to be clearer and updated it based on new results:

> "Leaves with small guard cells and a lower initial stomatal conductance (g_{i}) closed faster, but each explained variation in kinetic parameters at different levels of biological organization."

Comment R2.2 — Mechanistic follow-up on turgor-responsive window

This is really interesting and links back to early work on turgor maintenance and speed: "Smaller stomata can be speedier, but only if stomata are held at an aperture where they are responsive to changing turgor pressure." How could the authors mechanistically follow this up? Using anatomical measurements of g_{smax} as a benchmark for comparing to operational g_{sw} has some drawbacks — e.g. over-estimation. I think some of this is covered in the discussion.

Response:

We agree that more mechanistic follow-up studies are required. We added on lines 577-579 that:

> "Our study is limited by the fact that we did not directly measure guard cell turgor or stomatal aperture, but future studies should use newly developed methods (Brodersen *et al.* 2025) to directly evaluate the association between these traits."

Comment R2.3 — Wrong-way response magnitude and f_{gmax}

> The authors have chosen species with 'wrong way response' to VPD — do they take into account or explore the magnitude of the initial rise in g_{sw} and how this might relate to lag time or the f_{gmax} relationship (the magnitude of the 'hump' in Figure 1a)?

Response:

Unfortunately, as described in response to Comment R1.3, we did not log data on the magnitude of the WWR and therefore cannot analyze that with this data set. We agree that it would be useful to know whether the magnitude of the WWR affects the lag time or speed of closing. In the Discussion on lines 548-551, we added:

> "In addition to guard cell size, it will be useful to test whether the ratio of guard cell to adjacent epidermal size affects the strength of the WWR, which may be necessary to induce more rapid stomatal closure (Buckley *et al.* 2003; Pichaco *et al.* 2025)."

Comment R2.4 — Compensatory aperture in adaxial-blocked leaves (L403)

> L403 — stomatal aperture was more open in leaves where gas exchange through adaxial stomata was blocked — is this interesting, do the authors think it is compensatory?

Response:

We do not have any stronger data to support this idea, but we agree that this explanation makes sense. We added a comment about on lines 441-444:

> "This is presumably because a large fraction of stomata were blocked by the pseudohypo treatment and increased aperture of the abaxial stomata did not fully compensate. This observation differs from wheat, where g_{sw} on one surface was not affected by blocking gas exchange in the other (Wall *et al.* 2022)."

Comment R2.5 — Figure 1B legend format

> Fig 1b — Personal preference; although I appreciate the legend is correct, it might be clearer to make the large and small time constants a rectangle of the appropriate colour or dot of the colour? When I look at the figure I focus in on the format of the line rather than the colour which distracts from understanding the figure quickly.

Response:

We agree this improves clarity and have revised Figure 1b so that the legend for τ uses a solid color swatch rather than a line segment, visually distinguishing it from the line-style legend used for λ.

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